

Roll No.....

Mid Semester Examination

Name of the Program: **B.Tech(ECE)**

Semester: **III**

Name of the Course: **Signals and Systems**

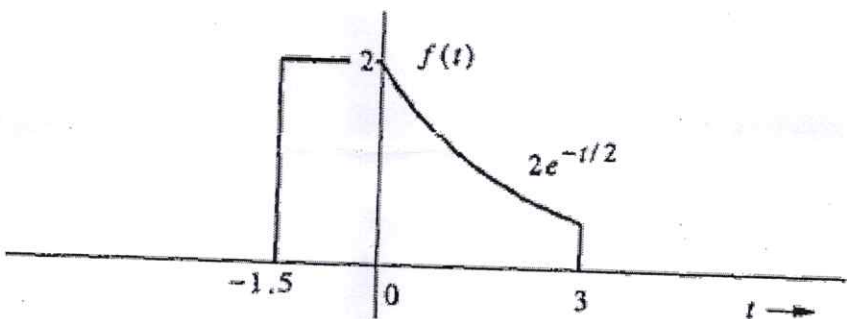
Course Code: **TEC 304**

Time: **90 minutes**

Maximum Marks: **50**

Note:

- Answer **all the questions** by choosing **any one of the sub questions**.
- Each question carries **10 marks**.

Q1	(10 marks)	CO1
(a) Determine whether or not each of the following signals are periodic. If the signal is periodic, determine its fundamental period. (i) $x[n] = \cos\left(\frac{\pi}{3}n\right) + \sin\left(\frac{\pi}{4}n\right)$ (ii) $x[n] = e^{\frac{j\pi n}{4}}$ (iii) $x(t) = \cos\left(t + \frac{\pi}{4}\right)$ (iv) $x(t) = \cos^2 t$		
OR		
(b) Figure shows a signal $f(t)$. Sketch and describe mathematically this signal time compressed by a factor 3. Repeat the problem for the same signal time expanded by factor 2.		
		
Q2	(10 marks)	CO1 & CO2
(a) Check whether the given system $y(t) = \text{Even}\{x(t)\}$ is (i) linear or nonlinear (ii) time invariant or time variant.		
OR		
(b) Determine the even and odd component of the following sequence: $x(n) = \{-2, 1, 2, -1, 3\}$ ↑		
Q3	(10 marks)	CO1 & CO2
(a) Evaluate the following integrals: (a) $\int_{-1}^1 (3t^2 + 1)\delta(t) dt$ (b) $\int_1^2 (3t^2 + 1)\delta(t) dt$ (c) $\int_{-\infty}^{\infty} (t^2 + \cos \pi t) \delta(t - 1) dt$		
OR		
(b) (i) A system has the input-output relation is given by $y[n] = \sum_{k=-\infty}^{n+1} x[k]$ Determine whether the system is (a) memoryless, or (b) Causal. (ii) Determine whether or not each of the following systems is invertible. If the		

	System is invertible, give the inverse system. (a) $y(t) = 2x(t)$ (b) $y[n] = \sum_{k=-\infty}^n x[k]$	
Q4	(10 marks)	CO2
(a)	Evaluate the following: (i) $\cos t \times \delta(t - \pi)$ (ii) $\int_{-\infty}^{\infty} e^{-t} \delta(2t - 2) dt$ (iii) $\int_{-\infty}^{\infty} e^{-t} \delta'(t) dt$.	
	OR	
(b)	Let $y(t) = x(t) * h(t)$. Then show that $x(t - t_1) * h(t - t_2) = y(t - t_1 - t_2)$	
Q5	(10 marks)	CO1 & CO2
(a)	What is LTI system and its impulse response? Explain how the impulse response is useful in finding out the LTI system output due to any arbitrary input signal.	
	OR	
(b)	Determine whether the following signals are energy signals, power signals, or neither (i) $x(t) = A \cos(\omega_0 t + \theta)$ (ii) $x(t) = u(t)$	